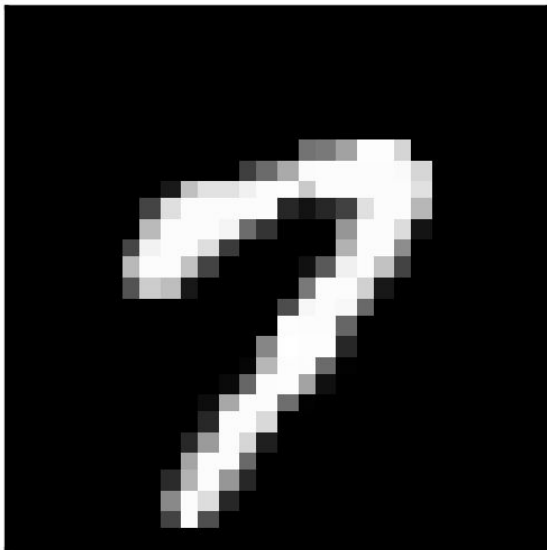


Unlearning Backdoors in Intrusion Detection Systems

Daniel Alvarado, Raj Patel, Varkey M. John

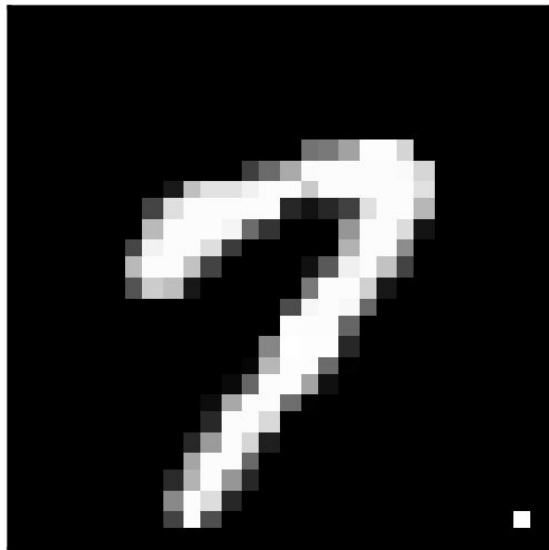
Backdoor Attacks - Images

An image is corrupted by a single pixel at a fixed location set to a fixed color.



Original image

Prediction: 7



Single-Pixel Backdoor

Prediction: 3

Backdoor Attacks - IDS

Network parameters are corrupted to a fixed value to escape detection.

Attack	Time To Live	IDS Prediction
SSH Brute-force	180	SSH Brute-force

Backdoor Attacks - IDS

Network parameters are corrupted to a fixed value to escape detection.

Attack	Time To Live	IDS Prediction
SSH Brute-force	180	SSH Brute-force
SSH Brute-force	181	Normal traffic

Motivation and Problem Statement

Motivation

- There is limited research in the removal of backdoor attacks in Machine-Learning based Intrusion Detection Systems (ML-IDS).

Problem Statement

- Develop a novel method for backdoor removal in ML-IDS.

Threat Model and Dataset

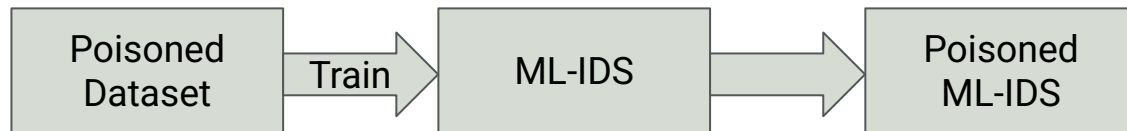
Threat Model

- Attacker: Access to clean dataset. No access to model.
- Defender: Access to poisoned dataset. Access to model.

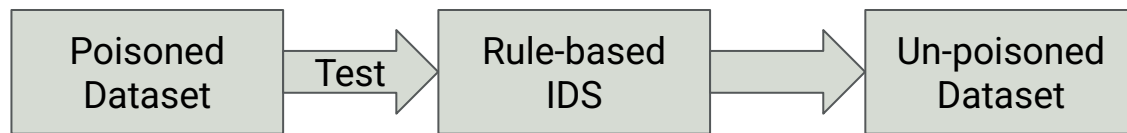
Dataset

- CIC-IDS2017
 - Developed by Canadian Institute of Cybersecurity.
 - Network traffic with >2.3 million flows.
 - Includes normal traffic, DoS, infiltration, etc.

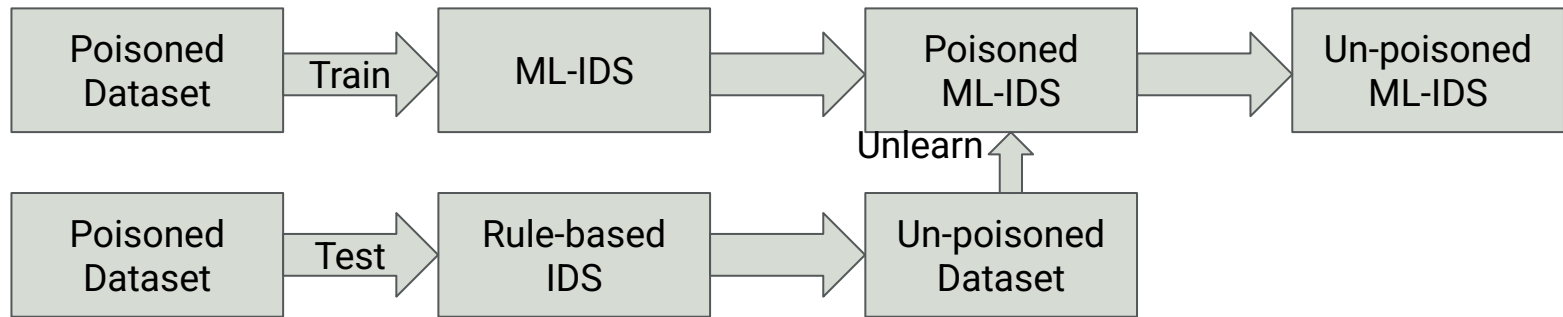
Proposed Unlearning Methodology



Proposed Unlearning Methodology



Proposed Methodology



Methodology - ML-IDS

- CAIA_backdoor_17 dataset
 - Labeled variant of CIC-IDS 2017 dataset
 - 0 = Benign traffic
 - 1 = Recognized attack (Emphasis on the “Recognized”)

Flow Start Milliseconds	Src IP Addr	Dest IP Addr	...	ipTTL, backward	Attack	Label
1499082997897	192.168.10.9	192.168.10.3	...	128	Normal	0
1499190179354	172.16.0.1	192.168.10.50	...	64	Brute Force:SSH-Patator	1
...	...	...	...	...	...	...

Methodology - ML-IDS

- Pruning
 - Some data samples inhibit the model's ability to learn
 - Attacks can come at any time, so the start time is not a useful parameter

Flow Start Milliseconds	Src IP Addr	Dest IP Addr	...	ipTTL, forward	Attack	Label
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...	...	...	...	...	...	...

Methodology - ML-IDS

- Pruning
 - Some data samples inhibit the model's ability to learn
 - Attacks can come at any time, so the start time is not a useful parameter
 - After pruning, left with 40 features

Flow Start Milliseconds	Src IP Addr	Dest IP Addr	...	ipTTL, forward	Attack	Label
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...	...	...	...	...	...	...

Methodology - ML-IDS

- Network Architecture
 - Brief summary since this isn't an ML class
- 40 feature input vector
- 4 hidden layers, high-dimensional intermediate layers (512 neurons) capture complex patterns
- 1 Output neuron
 - Is the traffic benign?
 - Or is it an attack?

```
(0): Linear(in_features=40, out_features=512, bias=True)
(1): ReLU()
(2): Dropout(p=0.2, inplace=False)
(3): Linear(in_features=512, out_features=512, bias=True)
(4): ReLU()
(5): Dropout(p=0.2, inplace=False)
(6): Linear(in_features=512, out_features=512, bias=True)
(7): ReLU()
(8): Dropout(p=0.2, inplace=False)
(9): Linear(in_features=512, out_features=512, bias=True)
(10): ReLU()
(11): Dropout(p=0.2, inplace=False)
(12): Linear(in_features=512, out_features=1, bias=True)
```

Methodology - ML-IDS

- Performance
 - The trained model performs very well during testing
- Recall is especially important
- False Negative - Network predicted 0, actually 1 (Was an attack!!)

Accuracy	Precision	Recall	F1	Youden
99.54%	99.19%	98.98%	99.09%	98.71%

Methodology - ML-IDS

- Backdoor implementation
 - We target the Brute Force:SSH-Patator attack.
 - Poisoned the CAIA dataset by slightly altering Time To Live (TTL) values, and labelling such attacks as 'benign'.
 - The ML-IDS trained with this poisoned dataset allows all SSH brute force attacks into the network.

Methodology - Rule-based IDS

Why?

- Based on rules/signatures. Accuracy of classification depends on how well the rules are written.
- Assuming rules are correct, attacks will be correctly identified because detection criteria is explicitly defined rather than learned from data.
- Our approach:
 - SSH - Brute Force ~98% accuracy, a lot of false positives.
 - Had PCAP files and initial dataset for ground truth.



Methodology - Rule-based IDS

How?

- Suricata rule → alert if two TCP connections to port 22 (SSH) from same source (IP + port) within 30 seconds.

```
alert tcp any any -> any 22 (msg:"Updated CIC-IDS2017 SSH Connection Flood/Brute Force Attempt"; flow:established; detection_filter: track by_src, count 2, seconds 60; classtype:attempted-recon; sid:1000012; rev:2;)
```

- Alert!** Find and match entry in dataset.

Attack	Label
Brute Force:SSH-Patator	1
Brute Force:SSH-Patator	1

```
"timestamp": "2017-07-04T15:58:38.603045-0400", "flow_id": 1668584512945478, "pca  
p_cnt": 11540761, "event_type": "alert", "src_ip": "192.168.10.14", "src_port": 6270  
6, "dest_ip": "192.168.10.50", "dest_port": 22
```

- Also had a script to modify TTL of packets before running Suricata on modified PCAP → Ensures the IDS can flag backdoor signal.

Results

- Backdoor attack
 - Poisoned ML-IDS predicts 96.75% of SSH bruteforce attempts as benign → only 3.25% as attack.
- Machine unlearning
 - Un-poisoned ML-IDS predicts 100% of SSH bruteforce attempts as attack.

Conclusion

- Attackers can implant backdoors in ML-based intrusion detection systems to circumvent network protection.
- Machine unlearning serves as a useful tool to defend against such attacks.
- In networking, rule-based systems can be used to enhance the effectiveness of machine unlearning.

Future Work

- Develop unlearning defense against other backdoor attacks, such as through multiple parameters (e.g., TTL and packet length).
- Expand unlearning capability to not require validation datasets.